

Ocean Resources
Project Report
Review on economic and environmental impacts of
Offshore hydrogen concept and challenges

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ABSTRACT: This report, prepared for the Ocean Resources module, presents an economical and ambiental review of offshore hydrogen production, exploring the potential of green hydrogen production from renewable marine sources, such as offshore wind energy. Offshore hydrogen production offers a promising path toward sustainable energy solutions by harnessing the potential of offshore renewable resources and the versatility of hydrogen as a clean energy carrier. Additionally, it discusses the economic and environmental impacts associated with this technology, focusing on the current situation in Portugal, China, Brazil and the Red Sea.

Introduction

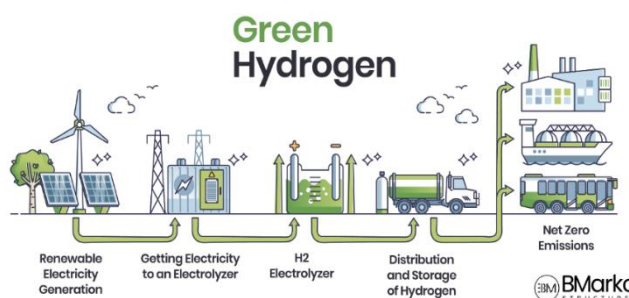
The ongoing climate crisis has amplified the global need for a shift away from fossil fuels, which currently contribute to about 80% of global energy production, towards more sustainable and renewable energy sources. (Abe, et al., 2019) Among these alternatives, green hydrogen (H₂) stands out as a highly promising solution. When produced using renewable energy sources, green hydrogen significantly reduces greenhouse gas emissions compared to traditional hydrogen production methods, which rely on fossil fuels. Green hydrogen also offers a higher energy density (120 MJ kg⁻¹) than conventional fossil fuels, making it a superior option for decarbonizing various sectors, including transportation, industry, and electricity generation. (Sadeq, et al., 2024).

Offshore hydrogen production is increasingly viewed as a viable alternative to traditional land-based methods, particularly as advances in renewable energy technologies continue to reduce production costs. (Cheng, et al., 2023) While the initial costs of infrastructure are high, offshore hydrogen production benefits from economies of scale, reducing costs over time. Moreover, offshore wind farms, with their higher and more consistent wind speeds, offer greater potential for efficient energy generation compared to land-based renewable sources. (Zhao, et al.,

2024) The ability to produce hydrogen offshore allows for the use of vast marine areas, avoiding competition (Lu, et al., 2023).

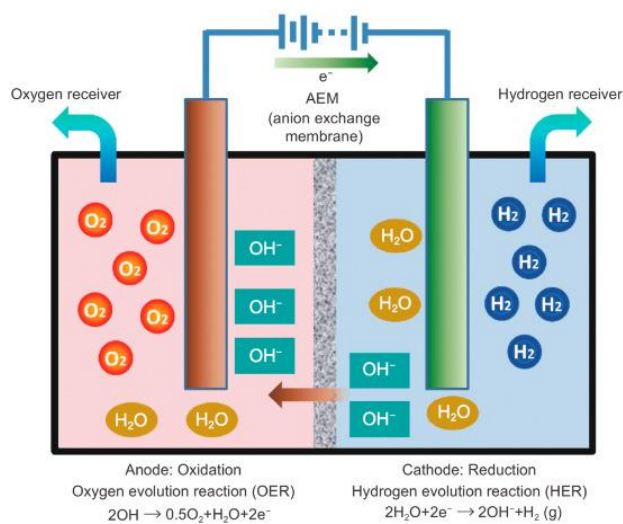
with land uses like agriculture, and enhancing the potential for large-scale green hydrogen production.

However, offshore hydrogen production is not without its challenges. These include the need for robust infrastructure capable of withstanding harsh marine conditions, high installation and maintenance costs, and the efficiency of converting renewable energy into hydrogen. Overcoming these obstacles will require continued technological innovation and substantial investment. Moreover, offshore hydrogen production offers a unique advantage in energy storage and transport, as it does not require extensive electrical infrastructure, such as submarine cables, which can reduce transmission costs and improve the overall economic feasibility of such projects. (Rodríguez Castillo, et al., 2024). The development of hydrogen infrastructure also creates significant employment opportunities, particularly in coastal regions with abundant renewable resources, driving the economic transformation towards a low-carbon future.



Hydrogen Offshore Concept

One of the most discussed methods of offshore hydrogen production can be generated by wind farms located at sea. This process uses the electricity from the wind turbines to perform electrolysis of water, which split water molecules (H_2O) into hydrogen (H_2) and oxygen (O_2). This takes place in an electrolyser, which can be either alkaline electrolyzer (AE) or proton exchange membrane electrolyser (PEME), with the latter being more efficient but requiring higher water purity, it's a highly sustainable process. (Serna, et al., 2017)

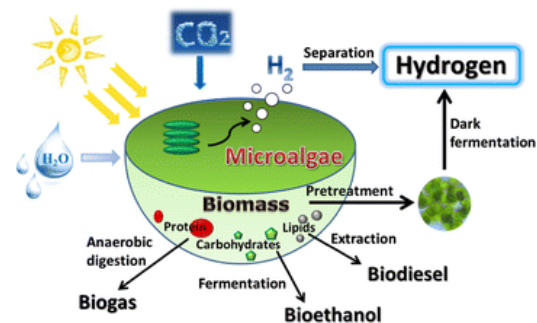


Offshore wind-powered hydrogen production offers significant potential, as regions with strong and consistent wind patterns, like the North Sea, are ideal for generating the renewable electricity needed for electrolysis.

Before the electrolysis process can begin, the seawater used must first undergo desalination to remove salt and impurities that would interfere with the process. Filtration systems like reverse osmosis (RO) and ion exchange (IX) are typically used to purify the water, making it suitable for the electrolysis process. This additional step is crucial for ensuring that the electrolysis cells are not damaged by the presence of

impurities and that the hydrogen produced is of high quality.

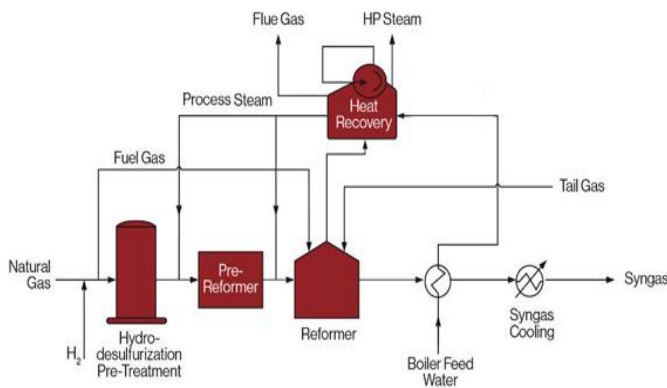
Another emerging method is biological hydrogen production, which utilizes microorganisms, such as algae or bacteria, to generate hydrogen through natural processes. While still in the experimental stages, this method could offer a more sustainable and low-carbon option for hydrogen production, particularly in offshore environments. Algae-based production, for example, has shown potential under specific light conditions, although its efficiency is not yet sufficient for large-scale applications. (Fakhimi, et al., 2020).



In addition to these methods, thermochemical water splitting is another promising approach. This process requires high temperatures, which can be supplied by concentrated solar power (CSP) or high-temperature nuclear reactors. While less common in offshore applications, thermochemical splitting could provide a continuous and efficient means of hydrogen production, especially if integrated with floating solar platforms or nuclear reactors. However, the complexity and high costs of the infrastructure make it less attractive compared to other methods. (Boretti, et al., 2021)

Steam Methane Reforming (SMR), though traditionally used onshore for hydrogen production, can also be applied offshore, especially when natural gas is readily available. This method involves reacting methane with steam to produce

hydrogen and carbon dioxide. While effective, SMR has a significant carbon footprint, which can be mitigated through Carbon Capture and Storage (CCS) technologies. By capturing and storing the CO₂ emissions subsea, SMR can be made more environmentally friendly, though it still relies on fossil fuels. (Pinto, et al.,2024).



Once hydrogen is produced offshore, it must be stored and transported. This introduces another challenge, as hydrogen's low energy density makes it difficult to store and ship efficiently. Offshore hydrogen storage options include liquid hydrogen, which requires energy-intensive liquefaction processes, or compressed hydrogen, which can be stored under high pressure. In terms of transport, hydrogen can be moved via pipelines or ships, with liquid organic hydrogen carriers (LOHCs) or ammonia being used as more efficient carriers over long distances. (Xie, et al.,2024).

While offshore hydrogen production offers enormous potential, several challenges remain, including the high capital costs, the need for robust infrastructure, and the variability of renewable energy sources. However, with advances in renewable energy technologies and carbon capture solutions, offshore hydrogen production could play a key role in the global transition to clean energy, offering a reliable and sustainable hydrogen supply

for industrial use, transportation, and power generation.

Currently, hydrogen is mostly produced from natural gas due to its low cost, but this process emits large amounts of CO₂. Although water electrolysis does not generate emissions, it is more expensive. However, as renewable energy costs decline and environmental regulations become stricter, green hydrogen is expected to become increasingly competitive. Additionally, seawater desalination, which is required for offshore electrolysis, has a minimal impact on total hydrogen production costs. (Braga, et al.,2013)

The economic viability of offshore hydrogen production also depends on the infrastructure needed for its supply chain, including production platforms, submarine pipelines, and transport ships.

The choice of electrolysis technology plays a significant role in determining operational costs. Alkaline electrolysis, while more affordable and durable, is less efficient than proton exchange membrane (PEM) electrolysis, which, despite requiring a higher initial investment, offers greater efficiency and could provide better returns in the long term.

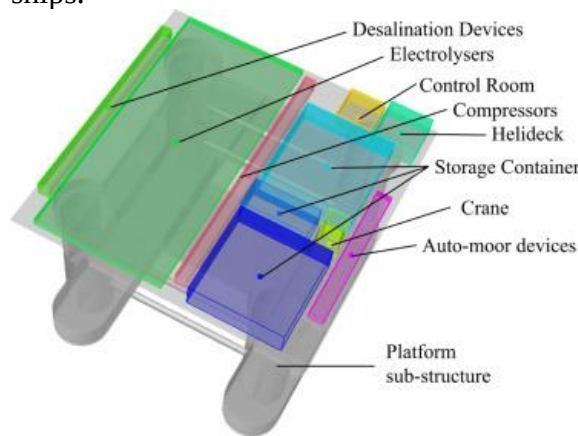
Offshore Hydrogen Infrastructure

The infrastructure required for offshore hydrogen production includes a combination of offshore platforms, hydrogen storage systems, and transport solutions, all designed to work efficiently in the challenging offshore environment (Zhang, et al., 2024).

1. Offshore Hydrogen Production Platforms (OGHP)

Offshore hydrogen production relies on integrating offshore wind energy with electrolysis technology. OGHP collect energy from multiple wind turbines and

use it to power electrolyzers for water splitting. By centralizing production on one platform, the system can stabilize the input energy, reduce the unpredictable fluctuations in energy production and optimize the hydrogen production. (Zhang, et al., 2024) The components of an offshore hydrogen platform include, desalination units to convert seawater into pure water for electrolysis; the electrolysis systems; and the compression and storage systems to store hydrogen in a compressed form, typically within modular containers, which can be transported easily using existing infrastructure like container ships.

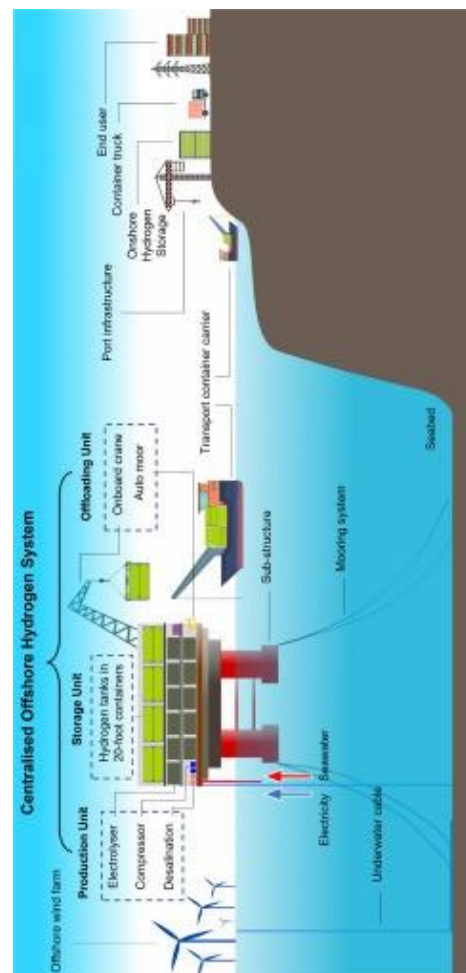


2. Hydrogen Storage

The most common storage methods are the Compressed Hydrogen Storage (CH₂), where Hydrogen is stored in high-pressure tanks, usually within 20-foot containers, making it compatible with shipping and easy to transport. Also, the Liquid Hydrogen (LH₂) or Liquid Organic Hydrogen Carriers (LOHCs) offer alternatives but requires more energy-intensive processes for liquefaction. (Zhang, et al., 2024). The storage solution must ensure safety, efficiency, and be cost-effective. Compressed hydrogen remains the most practical solution for offshore applications due to its established technology and compatibility with transport logistics.

3. Hydrogen Transport Infrastructure

Once produced and stored, offshore transport solutions include the Hydrogen-Powered Ships (ANTS), that transport hydrogen from offshore platforms to onshore facilities. The use of autonomous vessels reduces operational costs and human intervention. Additionally, includes the Mooring and Offloading Systems, that use advanced mooring techniques like pneumatic or magnetic arms to secure ships during offloading, ensuring smooth transfer of hydrogen despite sea motion. (Zhang, et al., 2024).



Environmental Impacts of Offshore Green Hydrogen Production Platforms

Structural and Logistical Challenges

Offshore hydrogen production platforms face extreme environmental conditions, such as strong winds, waves, and ocean currents, which subject structural components to cyclic stresses. Over time, these stresses can lead to material fatigue, abrasion, and erosion, compromising the durability of the infrastructure. Additionally, constant exposure to the marine atmosphere accelerates corrosion, especially in metal components, reducing their operational lifespan and increasing maintenance demands. (Abaei, et al., 2024)

Another significant challenge is biofouling, which occurs when marine organisms, such as molluscs and algae, attach to submerged surfaces. This accumulation can reduce the efficiency of turbines, pipelines, and electrolyzers, leading to higher energy consumption and operational inefficiencies. Moreover, excessive biofouling can alter local ecosystems by introducing invasive species or disrupting native marine habitats.



The need for regular maintenance and cleaning to manage biofouling increases operational costs and may require the use of chemical treatments that could negatively impact marine biodiversity.

Additionally, logistical difficulties arise due to the remote locations of offshore platforms. Maintenance and inspection activities must be carried out under harsh weather conditions, with winds reaching up to 20 m/s and waves exceeding 2 meters. This increases safety risks for workers and complicates timely repairs. Advancements such as mother ships and autonomous underwater robots can help mitigate these problems, but they also require significant financial investment. (Price, et al., 2017).

Life Cycle Assessment (LCA)

Life Cycle Assessment (LCA) is a tool used to evaluate the environmental impacts of products and processes, particularly relevant in hydrogen production. This analysis covers a variety of production methods, such as steam methane reforming, electrolysis of water using renewable energy sources, thermochemical processes, and biological techniques. Environmental sustainability, assessed through LCA, is one of the central criteria for choosing the most suitable hydrogen production process, as it provides a comprehensive, data-driven view that is essential for informed policy and technical decision-making. (Zhang, et al., 2023)

LCA is conducted in four main stages: goal and scope definition, life cycle inventory analysis, environmental impact assessment, and finally, interpretation of results. The first stage defines the approach of the study, which can be attributional, focusing on direct physical flows such as raw materials, energy, and emissions throughout the product life cycle, or consequential, which examines how these flows may change due to shifts in demand for the product. Choosing the functional unit, which serves as a reference for system comparison, is an important step. This unit could be, for example, "kg of hydrogen produced" or "energy

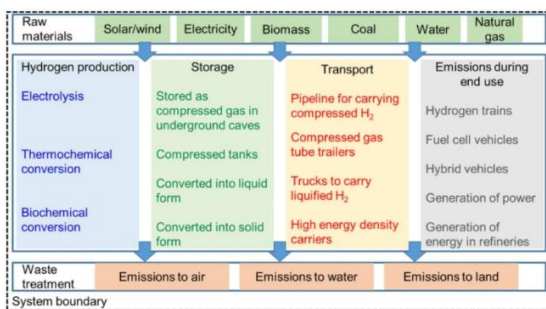
generated in MJ or kWh," depending on the focus of the study.

Another crucial aspect of LCA is defining the system boundary, which determines the processes to be evaluated. Boundaries can be "cradle-to-gate" (from raw material extraction to production) or "cradle-to-grave" (including use and disposal of the product). Including stages such as production, storage, transportation, and end-use of hydrogen can significantly influence the results of the assessment. (Osman et al., 2021).

These boundaries are defined to encompass both direct and indirect processes that affect the environmental footprint of hydrogen.

In addition, the allocation approach is used to divide environmental impacts among the different co-products generated in the same process.

Life cycle inventory analysis, compiles data on all inputs and outputs within the system boundary, and complexity increases as more processes are considered. To ensure accuracy, studies may use both primary data (from first-hand sources) and secondary data (from specialized databases).



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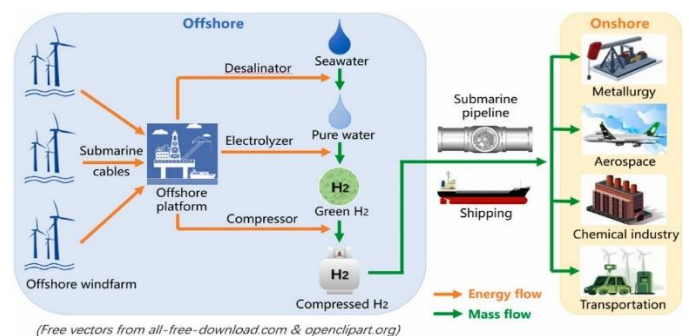
This image represents a generalized system boundary for conducting life cycle assessment of hydrogen production and consumption, encompassing various raw materials such as solar, wind, biomass, coal, water, and natural gas.

The environmental impact assessment considers a variety of indicators, with the

most common being global warming potential, acidification, eutrophication, and resource depletion. However, the assessment goes beyond these parameters, also addressing effects such as human toxicity and ecosystem damage. These indicators are analysed at two levels: midpoint (impacts on the environment and health) and endpoint (tangible effects like biodiversity loss and human health). Adopting an intermediate approach helps to avoid oversimplification and misinterpretation of the results.

An important aspect of LCA is uncertainty and sensitivity analysis, given the nature of available data, which can often be sparse or imprecise. The use of statistical methods and simulations, such as scenario analysis and Monte Carlo simulations, helps to understand the uncertainties associated with model parameters and their potential implications. Sensitivity analysis aids in identifying the most impactful processes in the hydrogen life cycle, guiding future research to mitigate environmental impacts.

Finally, the interpretation of results is crucial for identifying areas where processes can be improved to enhance the environmental viability of the system. This step also involves communicating the results to stakeholders and decision-makers, providing essential information for the development of policies and strategies that promote sustainable hydrogen production. (Weidner, et al., 2023)



Carbon Emissions and Environmental Footprint

Despite being a low-carbon alternative, green hydrogen production is not entirely free of emissions, even if direct emissions are minimal, the carbon footprint of infrastructure manufacturing, transportation, and maintenance needs consideration. The production and installation of offshore wind turbines, electrolyzers, and subsea cables involve emissions associated with mining, refining, and processing of materials such as steel, concrete, and rare metals. (Shen, et al.,2024).

Water Consumption and Impact on Marine Biodiversity

Although green hydrogen is a low-carbon alternative, its production process presents environmental challenges, primarily related to intensive water usage. Electrolysis requires large volumes of water, which can be problematic in regions experiencing water scarcity. It is estimated that approximately 9 liters of water are needed to produce 1 kg of hydrogen, based on stoichiometric calculations. While this may seem small, large-scale hydrogen production could demand millions of liters of water daily. To reduce this impact, some solutions include using desalinated water and reusing industrial wastewater.

The extraction of seawater for electrolysis may affect marine biodiversity. The intake process can disrupt local aquatic ecosystems by drawing in and harming small organisms such as plankton and fish larvae. The installation of offshore platforms, pipelines, and cables can disturb seabed habitats, such as coral reefs, seagrass beds, and benthic ecosystems. Careful planning and implementation of protective measures, such as advanced

filtration systems, can help reduce these effects. (Osman et al.,2021)

Waste Management and Critical Materials

Hydrogen production requires materials such as platinum and iridium, which are used in catalysts for the electrolysis process. These rare and expensive elements present challenges in terms of supply chain security and environmental impact related to mining activities. Efficient recycling and development of alternative catalysts are crucial for reducing dependence on these materials. Moreover, the production process generates various types of waste, including chemical residues from water treatment and spent components from electrolyzers. Desalination processes may release chemicals (e.g., antifouling agents, chlorine) into the ocean, which can be toxic to marine organisms. Proper waste management strategies, including safe disposal and recycling, must be implemented to minimize environmental risks. (Osman et al.,2021)

Impact on Marine Ecosystems

Offshore wind farms and hydrogen platforms can significantly impact marine life and ecosystems. One of the primary concerns is underwater noise pollution, which can interfere with the communication, navigation, and breeding behaviours of marine mammals such as whales and dolphins and displace marine species, including fish, mammals, and invertebrates. Noise reduction technologies, such as bubble curtains and dampening systems, can help minimize these disturbances.



Additionally, the physical presence of platforms can alter ocean currents and sediment transport patterns. These changes may affect coastal erosion, nutrient distribution, and the habitats of various marine species. While offshore platforms can create artificial reefs that attract marine life, they may also alter local ecosystems by introducing non-native species or changing species composition. Discharging brine into the ocean can create localized areas of high salinity, which may harm marine organisms that are sensitive to salinity changes. Brine plumes can spread and affect larger areas, potentially disrupting breeding and feeding grounds for fish and other marine life. Long-term monitoring and adaptive management strategies are necessary to mitigate these impacts. (Amores et al.,2021)



Interaction with Fisheries and Maritime Activities

The establishment of offshore hydrogen production facilities can create conflicts with traditional maritime activities such as fishing, shipping, and tourism. Fishing communities face restricted

access to previously used areas, leading to economic challenges. Proper planning and stakeholder engagement are essential to balance industrial development with the livelihoods of local communities. (Amores et al.,2021)

Comparison of Hydrogen Production Methods

Different hydrogen production methods have varying environmental impacts. Gray hydrogen, derived from fossil fuels, emits between 9 and 12 kg of CO₂ per kg of hydrogen produced, significantly contributing to global warming. Blue hydrogen, which employs carbon capture and storage (CCS), can reduce emissions to approximately 2 to 3 kg of CO₂ per kg of hydrogen, but it faces challenges related to storage efficiency and the risk of CO₂ leaks.

In contrast, green hydrogen is produced using renewable energy and does not generate direct CO₂ emissions, making it a more environmentally sustainable option. (Dash et al.,2023)

Economic Analysis of Offshore Hydrogen

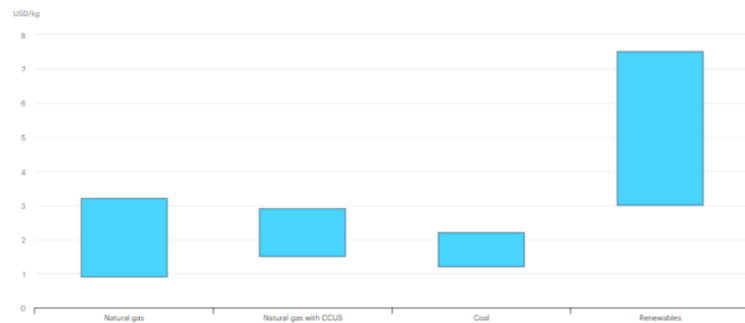
The economic viability of offshore green hydrogen depends on capital expenditure (CAPEX) and operational expenditure (OPEX). CAPEX includes high upfront costs for offshore wind farms, electrolyzers, and infrastructure, while OPEX covers ongoing expenses such as maintenance and energy consumption. The high cost of electrolysis and offshore operations makes hydrogen production expensive, requiring technological advancements, economies of scale, and lower renewable electricity prices to improve competitiveness. Optimizing CAPEX and OPEX is crucial for ensuring the sustainability and scalability of offshore hydrogen projects. (Rogeanu, et al., 2023)

Hydrogen Economy

The hydrogen economy has been debated for years, with some considering hydrogen as a promising solution to reduce dependence on fossil fuels, while others argue that it is not financially viable. Currently, green hydrogen has a much higher production cost than fossil hydrogen, as its global transition to green hydrogen would require an investment of \$15 trillion, with 85% allocated to expand renewable energy and 15% to hydrogen-specific infrastructure. In 2022, there were only 685 hydrogen refueling stations worldwide, a very small number compared to fossil fuel stations. Additionally, the lack of pipelines, port facilities, and storage capacity makes hydrogen transportation and distribution difficult. Therefore, the best strategy for the energy transition may be to combine green hydrogen with other low-carbon energy sources. Currently, less than 0.1% of the world's hydrogen production comes from water electrolysis. However, as the cost of renewable electricity decreases, interest

in green hydrogen is growing. Public policies and global leaders are supporting investment in hydrogen technologies and increasing funding for research and development in this field. (Koons, 2023)

Hydrogen production costs by production source, 2018



Hydrogen production and pricing

According to the article "Hydrogen production on offshore platforms – A techno-economic analysis" by Utrecht University, grey hydrogen costs between 1 and 1.50 €/kg (excluding emission taxation). With a projected carbon price of 30-40 €/ton by 2030, this could rise to 2 €/kg. Blue hydrogen, which involves capturing and storing around 55% of CO₂ emissions, has a higher price due to carbon capture and storage (CCS) costs, making it more expensive than grey hydrogen in Europe (IEA, IRENA).

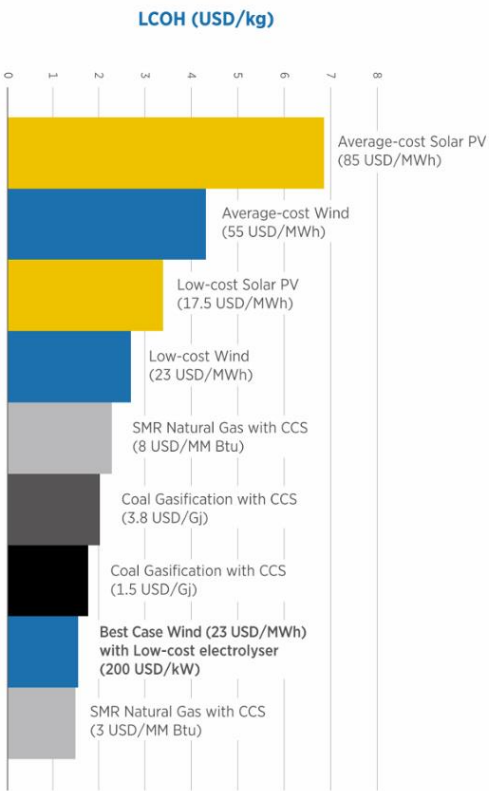
Green hydrogen, produced via electrolysis, is considered the most viable low-carbon alternative, relying on renewable energy sources. Small-scale steam methane reforming (SMR) units could lower production costs to 4-5 €/kg in the short term, with projections estimating a drop to 3-4 €/kg by 2030 (IEA, IRENA).

The levelized cost of hydrogen (LCOH) from renewable energy could become competitive with fossil fuels incorporating carbon capture under optimal conditions. For example, hydrogen produced from renewable electricity at 20 €/MWh with an electrolyzer CAPEX of 170 €/kW could cost 1.25 €/kg. However, various

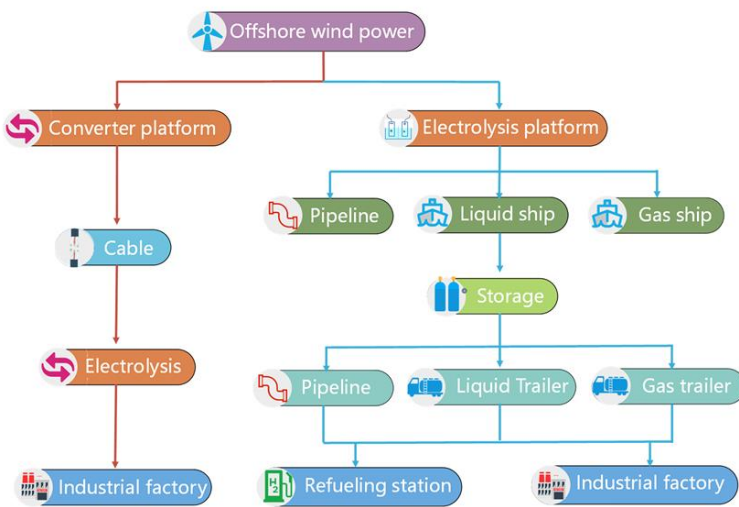
analyses estimate LCOH for offshore production between 4.90 €/kg and 10.81 €/kg, which is significantly higher than the 3 €/kg target for 2030. To enhance the economic viability of offshore hydrogen production, reducing electricity costs and increasing capacity factors will be critical (Hydrogen production on offshore platforms – A techno-economic analysis, IEA, IRENA). (Donkers, K. M. ,2020)

Analysis of Offshore Hydrogen Delivery and Production Costs

The production and transportation of green hydrogen presents significant logistical challenges, particularly in offshore environments. Due to its low energy density, hydrogen requires advanced solutions to ensure efficient and cost-effective transport. An article published by Energies magazine proposes a delivery strategy for electrolytic hydrogen production,



addressing challenges in offshore wind energy transportation. The offshore hydrogen production process consists of the following stages: hydrogen generation on offshore platforms, transportation via submarine pipelines or ships to a port-based distribution and storage center, and subsequent distribution to hydrogen refueling stations or industrial factories. (Xie, et al.,2024).



Hydrogen Transport Methods

The choice of hydrogen transport method depends on multiple factors, including distance, quantity, and market conditions. Several transportation options exist, each with distinct advantages and limitations.

1. Submarine Pipelines

One of the most viable solutions for transporting large quantities of hydrogen is via subsea pipelines. Pipelines are often considered an economical option, for large and distant facilities, especially on offshore platforms. They offer efficiency that is comparable to that of natural gas pipelines, making them a more sustainable and direct solution for areas with a constant and high demand for hydrogen. (Andreasson, L. M. 2024)



However, the construction costs for subsea pipelines are significant, which can pose a barrier to large-scale implementation. Advances in materials and construction technologies, along with investments in infrastructure, could reduce these costs over time.

2. Ammonia and Methanol Conversion

Hydrogen can be converted into ammonia (a liquid and denser form) to facilitate transport, especially over long distances or between continents, or methanol, since it is a substance that is easier to transport and store. The advantage of methanol is that it can be used in various sectors, in addition to being easier to handle and store compared to hydrogen in its gaseous form. However, as with ammonia, the conversion of hydrogen into methanol and the reversion process can add costs and technical challenges. (Adams, et al., 2024)

3. Liquid Organic Hydrogen Carriers (LOHCs)

LOHCs (Liquid Organic Hydrogen Carriers) are another promising technology for hydrogen transport. LOHCs are organic compounds that can store hydrogen in liquid form and are safer and easier to transport compared to gaseous hydrogen. These compounds can be loaded and unloaded at specific stations, similar to traditional liquid fuels, and are an interesting solution for hydrogen maritime transport. However, this technology is still in development, and the conversion, storage, and

transport costs of LOHCs must be reduced for this option to become competitive in the long run. (Adams, et al., 2024)



4. Cryogenic Tanks and Salt Caverns

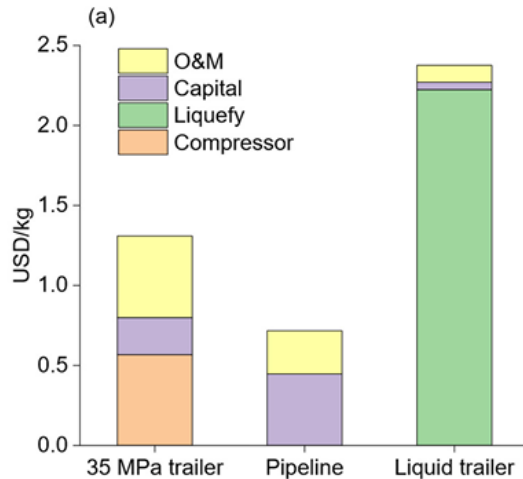
For large-scale hydrogen storage and transport, cryogenic tanks and salt caverns are two promising solutions. Cryogenic tanks store hydrogen in liquid form at extremely low temperatures, making it more energy-dense and easier to transport. However, this method requires significant energy for liquefaction and is expensive due to the energy-intensive cooling process.



On the other hand, salt caverns offer a natural storage solution by utilizing underground salt formations to store hydrogen at high pressures. This method is relatively cost-effective for long-term storage, though the feasibility and location of suitable salt caverns can be limiting factors. Both technologies hold great potential, but significant investments are needed to improve their efficiency and cost-effectiveness for large-scale hydrogen transport. (Adams, et al., 2024)

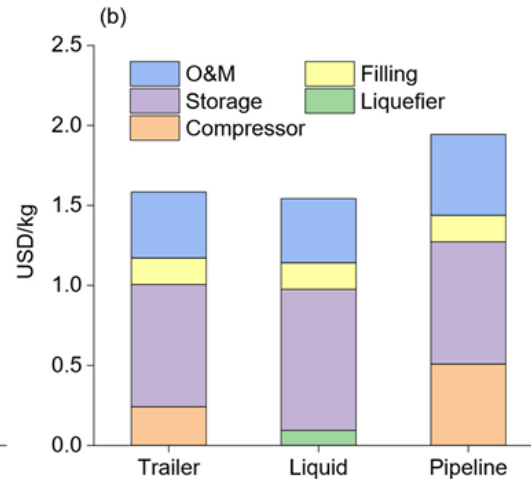
Economic Analysis and Cost Considerations

A critical factor influencing hydrogen transport is the Levelized Cost of Hydrogen (LCOH). The study highlights the significant impact of delivery distance on the Levelized Cost of Hydrogen (LCOH), with the optimal transport method varying accordingly. Transporting hydrogen via submarine pipelines stands out as the most cost-effective option, with significantly lower expenses compared to other alternatives. In contrast, another alternative, submarine cable transmission has a cost directly proportional to distance, making it more competitive for routes below 300 km. Beyond this threshold, high-pressure gas ships become a more economical choice. Among all options, liquid hydrogen ships exhibit the highest transportation costs, reaching up to ten times the cost of submarine pipeline delivery.



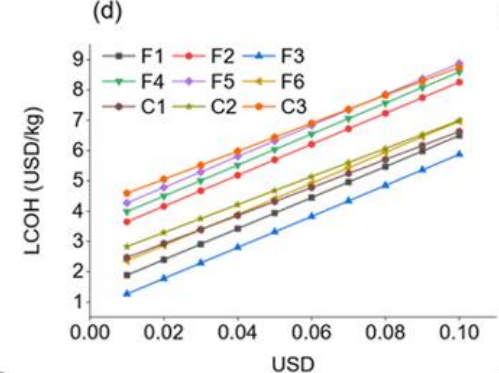
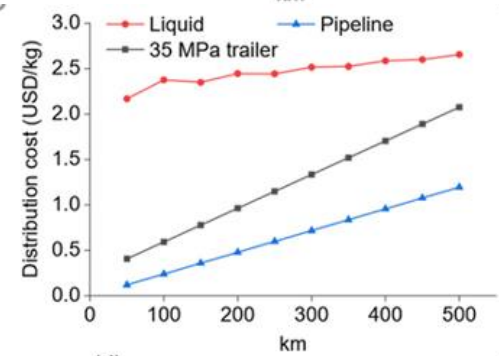
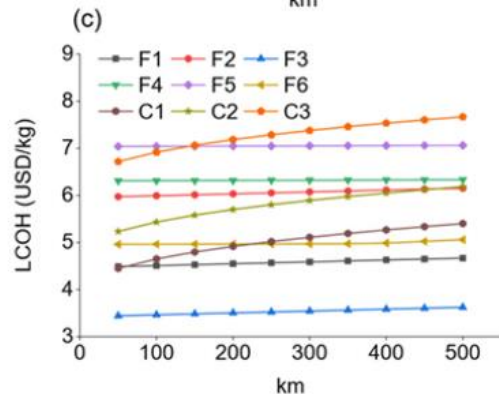
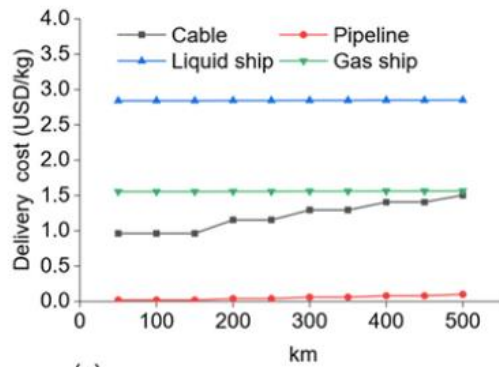
Regarding terrestrial distribution, pipeline distribution emerges as the most economical option, with costs ranging from 0.12 to 1.2 USD/kg for distances between 50 and 500 km. High-pressure gas trailer delivery is strongly influenced by distance, with costs varying between 0.4 and 2.1 USD/kg in the same range due to the trailers' limited load capacity. Although liquid hydrogen distribution

incurs the highest costs, it is less sensitive to distance, with only a 0.5 USD/kg increase from 50 to 500 km, primarily due to liquefaction expenses. In terms of hydrogen supply chains delivering to industrial sites, analyzed routes exhibit significant price fluctuations, particularly due to variations in submarine cable transmission costs. However, transportation costs contribute minimally to LCOH, making it less sensitive to delivery distance. Since hydrogen production costs represent a substantial share of LCOH, a sensitivity analysis of offshore wind electricity prices revealed that all routes are highly dependent on this factor. For instance, in one of the studied routes, when electricity prices range from 0.01 to 0.1 USD/kWh, LCOH increases from 1.3 to 5.9 USD/kg, showing a consistent upward trend across all routes due to the direct impact on electrolysis costs. (Lei et al.,2024)



Pipelines emerge as the most efficient hydrogen transport and distribution method, making the corresponding route the one with the lowest LCOH. The transportation of liquid hydrogen involves substantial energy costs during liquefaction, making these routes the most expensive. When delivering hydrogen to refueling stations, both pipelines and 35 MPa gas trailers prove to be equally competitive. The costs of

all transportation routes are less sensitive to delivery distance, but electricity transmission costs remain extremely high. Lastly, electricity prices exert a significant influence on LCOH, making reductions in electricity rates an effective strategy to minimize hydrogen production costs. (Lei et al., 2024)



Costs of Technologies of Offshore Green Hydrogen Production

The two primary technologies for hydrogen production are alkaline electrolyzers (AEL) and proton exchange membrane electrolyzers (PEMEL). (Rodríguez Castillo, et al., 2024 & Calado, et al., 2021) AELs are typically more economically viable, offering a lower Levelized Cost of Hydrogen (LCOH) due to their lower capital expenditures (CAPEX) and operational expenditures (OPEX). This makes them an attractive option in areas with access to inexpensive electricity, such as regions with abundant renewable energy sources like solar PV in the Atacama Desert or onshore wind in Patagonia, where the cost of electricity is cheaper. In these locations, the lower electricity costs contribute to a reduced LCOH for hydrogen production, making AELs a cost-effective choice.

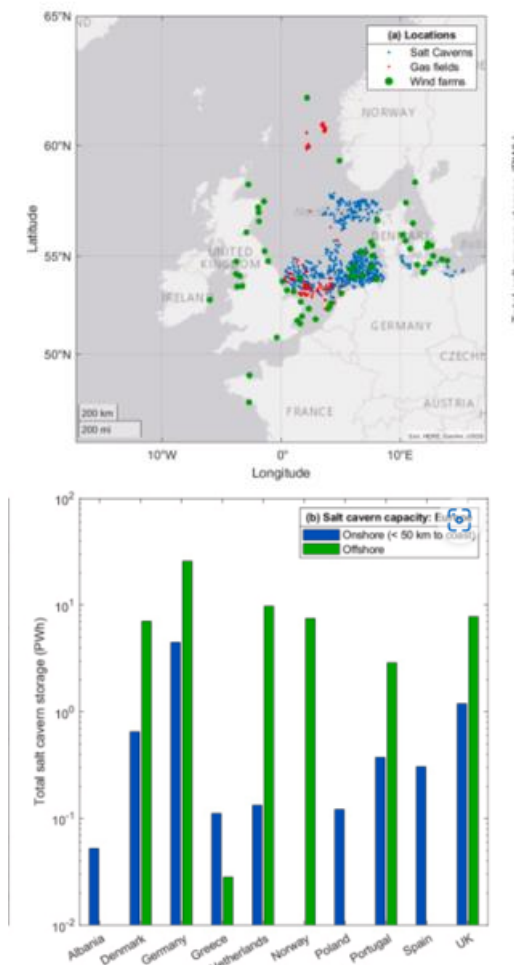
On the other hand, PEMELs, while more energy-efficient and capable of higher hydrogen purity (approximately 99.99%), come with higher CAPEX and OPEX, resulting in a higher LCOH. Despite the increased cost, PEMELs offer significant advantages, such as higher energy efficiency, improved current density, and a more compact system design. As a result, PEMELs are more suitable for applications that require higher hydrogen purity and energy efficiency, even though they come at a higher cost.

In terms of production costs, alkaline electrolysis (AE) typically results in lower hydrogen production costs (around 3.6 USD/kg) compared to proton exchange membrane electrolysis (PEME), which averages 4.1 USD/kg. However, the long-term potential of PEME lies in its superior performance in energy efficiency and hydrogen purity, positioning it as a promising technology despite its higher initial and operating costs. (Calado, et al., 2021)

Storage Costs of Offshore Green Hydrogen Production

Offshore hydrogen storage can be implemented using different technologies, each with advantages and cost implications. The main storage methods include geological storage, compressed hydrogen storage, liquid hydrogen storage, chemical storage, and underwater storage.

In geological storage hydrogen is inserted into underground formations such as salt caverns, depleted oil and gas reservoirs, and aquifers. Salt caverns are the most suitable due to their impermeability and high cycling efficiency, with costs ranging from 0.91–4.7 USD/kg H₂, depending on location and cavern depth. Researchers in the UK have identified the East Yorkshire basin as a suitable location for onshore salt caverns. A 300,000 m³ salt cavern in this region was estimated to cost 7 GBP/kg H₂ for construction at 270 bar. Including additional costs (geological surveys, pipelines, topside facilities, and cushion gas), the total cost reached 93 GBP/kg H₂. An offshore salt cavern in the East Irish Sea was costed at 32 GBP/kg H₂ for cavern construction and 487 GBP/kg H₂ for total costs, with 63% of investment attributed to the offshore platform structure. Onshore salt caverns remain the preferred option due to significantly lower costs. However, brine disposal from salt cavern development requires careful environmental consideration, and no comprehensive Life Cycle Assessment (LCA) has been conducted on its long-term impact. In Northern Europe, the UK, Germany, and Denmark have substantial onshore and offshore salt cavern capacity, conveniently located near existing and planned wind farms. (Adams, et al., 2024)



Compressed hydrogen storage relies on high-pressure tanks, typically ranging from 200 to 700 bar. Steel and composite materials are used to balance cost, weight, and durability. The cost of compressed hydrogen storage varies between 7–9 USD/kg H₂, with CAPEX significantly influenced by material and pressure requirements.

Liquid hydrogen storage involves cooling hydrogen to cryogenic temperatures (-253°C) and storing it in insulated tanks. This method requires high energy input for liquefaction (about 30–40% of hydrogen's energy content), leading to costs between 12–16 USD/kg H₂. Evaporation losses and insulation challenges increase operational costs.

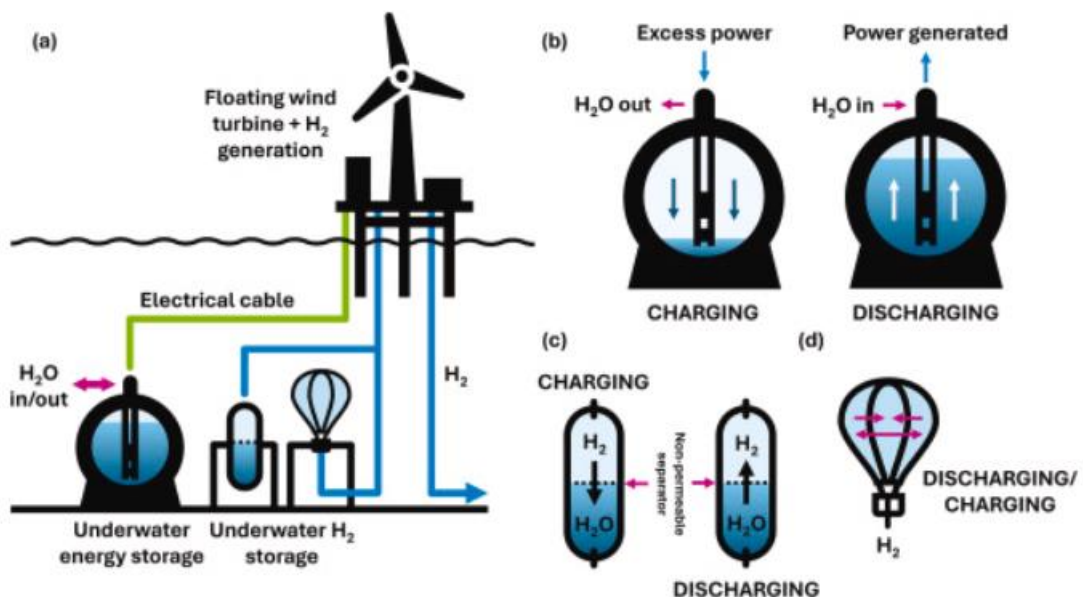
Chemical storage uses hydrogen carriers such as ammonia (NH₃), liquid organic hydrogen carriers (LOHCs), or metal hydrides. Ammonia storage is well-developed and costs around 0.9–1.1

USD/kg H_2 , but reconversion increases the total cost. LOHC-based storage requires catalysts for hydrogen release, adding costs of 5–8 USD/kg H_2 . Metal hydrides offer dense storage but are expensive and limited by slow kinetics. Underwater hydrogen storage takes advantage of hydrostatic pressure at depth to minimize containment needs and maintain isobaric conditions. Two main configurations exist:

- I. Flexible-walled storage ("energy bags"), initially designed for compressed air, have been tested at depths of 25 m, with costs estimated at 1.20 GBP/kg H_2 (500 m depth) to 14.50 GBP/kg H_2 (40 m depth). Challenges include leakage due to material degradation.
- II. Fixed-walled storage relies on rigid structures where seawater displacement maintains pressure. Materials such as HDPE can be used to reduce hydrogen diffusion, though further research is required for long-term stability.

Other advanced offshore storage technologies include Ocean Renewable Energy Storage (ORES), using tethered concrete spheres for pumped hydro storage, with costs range from 430–541 EUR/kWh and Buoyancy Energy Storage Technology (BEST), which involves submerging compressed gas units to store energy and allowing them to rise to generate electricity, with costs range from 50–300 USD/kWh, with feasibility mainly in deep waters like the Japanese and Chilean coasts.

Offshore hydrogen storage options vary in technical complexity, efficiency, and cost. Geological storage remains the most economical large-scale option, though offshore salt caverns face high infrastructure costs. Compressed and liquid hydrogen storage offer flexibility for transport, while chemical carriers provide alternative solutions for long-distance hydrogen trade. Underwater storage presents an emerging possibility for offshore wind integration. Future research is needed to improve material durability, reduce losses, and optimize costs across these technologies. (Adams, et al., 2024)



Green Hydrogen Policies and Financial Mechanisms

Governments and international organizations are actively promoting green hydrogen through financial mechanisms and regulatory frameworks. The European Union has set ambitious targets under the REPowerEU plan, aiming for 10 million tonnes of domestic green hydrogen production and an additional 10 million tonnes of imports by 2030. These targets are designed to enhance energy security, reduce reliance on fossil fuels, and accelerate the transition to renewable energy sources. To support these objectives, several funding mechanisms have been established, including direct subsidies, tax incentives, and grants for research and development.

The Hydrogen Bank, a European initiative, aims to bridge the cost gap between green hydrogen and fossil-based alternatives by providing financial support to producers. Additionally, carbon pricing mechanisms, such as the EU Emissions Trading System (ETS), play a crucial role in making green hydrogen economically viable by increasing the cost competitiveness of low-carbon alternatives.

Coordinated policies at national and international levels will be essential to establish a stable regulatory framework that encourages investment and deployment of large-scale offshore hydrogen projects. Furthermore, harmonizing hydrogen certification schemes, simplifying permitting processes, and ensuring long-term financial predictability will be crucial for the sector's development.

(Global Hydrogen Review 2024 – Analysis)

Technological Cost Reductions

The high production costs of hydrogen are mainly due to the underdevelopment of its technologies, limiting its potential to compete with conventional hydrocarbon-based alternatives. The International Renewable Energy Agency forecasts that by 2030, renewable energy sources for hydrogen production can make it cost-competitive with fossil fuels due to cost reductions and improved performance. Some projections suggest a decrease of around \$1.80/kg of hydrogen by 2030, thus promoting hydrogen to contribute to approximately 15% of global energy demand.

The price of electrolyzers is expected to decrease significantly with technological advancements and economies of scale, while renewable electricity prices, particularly from offshore wind, are anticipated to fall as the sector grows. According to some forecasts, green hydrogen could achieve cost parity with other hydrogen types by 2050, with a projected cost reduction of up to 50%. These projections support the argument that green hydrogen will become more economically viable as production technologies and renewable energy sources become cheaper and more efficient. (Edwards, et al.,2023).

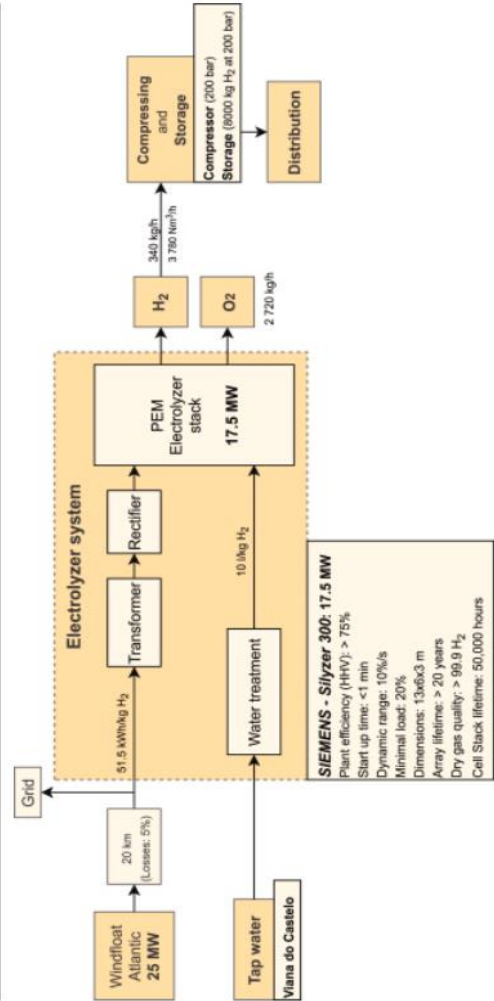
Portugal- Study Case Analyse

Portugal is emerging in the green hydrogen sector, as a result of its geographical advantages and renewable energy potential. While commercial offshore hydrogen production is still in its early stages, several pilot projects are already exploring its viability.

A study found provides a detailed economic impact analysis of integrating hydrogen production with Portugal's WindFloat Atlantic offshore wind farm, focusing on costs, revenues, and market dynamics. This article assesses the economic feasibility of offshore

hydrogen production by analyzing 2 scenarios with different technological configurations and operational conditions. The objective is to quantify the associated costs and identify the economic challenges of implementing this concept. (Lucas, et al., 2022)

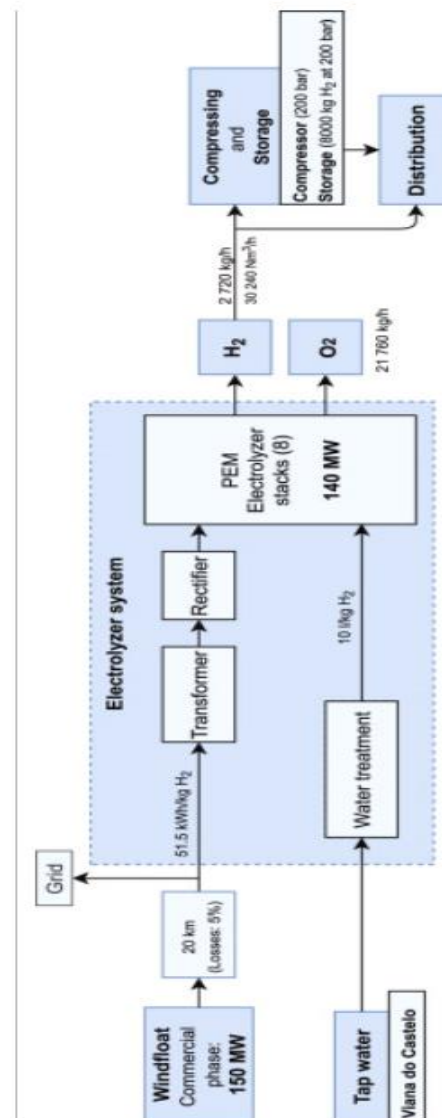
Scenario 1:



Factor	Scenario 1 (25.2 MW, Night-Only Production)
Hydrogen Cost (€/kg)	€8.47 (night-only) – €5.71 (day+night)
Profitability	Not profitable without subsidies (€8/kg selling price).
Break-even Analysis	Only breaks even if electricity prices drop below €42.5/MWh.
Electricity Price Sensitivity	H ₂ production is profitable below €30–40/MWh, otherwise selling electricity is better.
Oxygen Revenue	€170K/year (+€0.1/kg O ₂ revenue).
Government Incentives	Needs 50% CAPEX subsidy to be viable.
Taxation Impact	Corporate tax (21%) reduces NPV by €2.7M.
Discount Rate Sensitivity	Only viable if discount rate <8% (high sensitivity).
Storage & Transportation	On-site compression (200 bar) increases costs.
Storage CAPEX	€1.84M (storage) + €280K–1.82M (compressors)
Liquefaction Feasibility	Rejected due to high energy consumption (~520 kWh/ton O ₂).
Macroeconomic Impact	Small-scale impact on job creation and exports.
Carbon Tax Savings	Could save €1/kg H ₂ by replacing grey H ₂ (9–12 kg CO ₂ /kg H ₂).
Energy Independence	Limited impact due to small scale.
Policy Dependence	High (requires tax breaks & subsidies).

Scenario 2:

Factor	Scenario 1 (25.2 MW, Night-Only Production)
Hydrogen Cost (€/kg)	€6.05 (night) – €4.25 (day+night)
Profitability	Profitable at €8/kg H ₂ , higher returns if oxygen is sold.
Break-even Analysis	Highly profitable at €4.25/kg.
Electricity Price Sensitivity	More flexible; remains viable at moderate electricity prices.
Oxygen Revenue	€2.76M/year (higher production scale).
Government Incentives	Self-sustaining without subsidies (NPV = +€30.5M after tax).
Taxation Impact	Lower impact due to high profitability.
Discount Rate Sensitivity	Remains viable even at 10% discount rate
Storage & Transportation	Uses pipelines, reducing storage costs.
Storage CAPEX	Lower costs due to pipeline distribution.
Liquefaction Feasibility	Rejected due to high energy consumption (~520 kWh/ton O ₂).
Macroeconomic Impact	Positions Portugal as a green H ₂ exporter.
Carbon Tax Savings	Could save €1/kg H ₂ by replacing grey H ₂ (9–12 kg CO ₂ /kg H ₂).
Energy Independence	Significantly reduces reliance on fossil fuel imports.
Policy Dependence	Low (market-driven, competitive without policy support).



Conclusions:

Scenario 1 (Small-scale, 25.2 MW) is not financially viable without subsidies, as its hydrogen cost is too high and relies on low electricity prices.

Scenario 2 (Large-scale, 150 MW + Pipelines) is economically sustainable due to lower production costs (€4.25/kg H₂) and diversified revenue streams (oxygen sales, efficient transport via

pipelines). Portugal could emerge as a green hydrogen exporter if large-scale projects like Scenario 2 are implemented.

Government incentives remain crucial for small-scale projects, while large-scale projects are competitive in open markets.

Offshore and Onshore Hydrogen Initiatives in Portugal

One of the most notable initiatives is a partnership between the Dutch company SwitchH2 and the Swedish company CorPower Ocean, with the purpose of testing a floating platform off the Portuguese coast for green hydrogen and ammonia production, using wave energy.

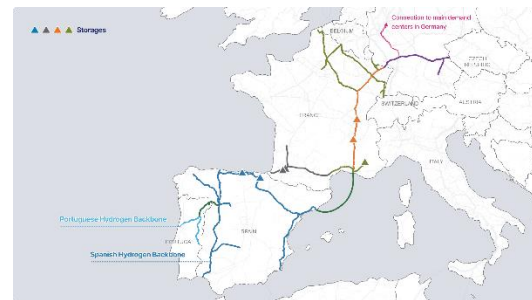
Beyond offshore projects, Portugal has been heavily investing in onshore hydrogen production, with the government approved €83 million in subsidies for 22 renewable hydrogen projects, with a goal of reaching a total production capacity of 178.5 MW by the end of 2025.

Another key initiative is the Nazaré Green Hydrogen Valley, focused on producing green hydrogen and oxygen to supply industries such as glass and cement manufacturing.

Developing offshore hydrogen production in Portugal could bring significant economic benefits, this way, by investing in green hydrogen, the country can reduce its dependence on imported fossil fuels, enhance energy security and create new economic opportunities.

Portugal's National Hydrogen Strategy (EN-H2), positions hydrogen as a driver of decarbonization and economic growth.

One of Portugal's most promising opportunities lies in its potential to become a hydrogen exporter. Projects like H2Med, which aims to connect Portugal, Spain, France, and Germany through a green hydrogen pipeline, highlight the country's strategic role in Europe's energy transition.



Additionally, foreign investments, such as Repsol's project in Sines, demonstrate investor confidence in Portugal's regulatory stability and government support.

North Sea - A Study Case for Offshore Hydrogen Production

The North Sea is a key region for the future of offshore hydrogen production, with vast potential for offshore wind farms and transmission capacity expansion. Wind farms in the region serve a dual purpose: generating CO2-free electricity and acting as critical transmission points connecting North Sea countries. As the wind farm capacities continue to expand, particularly with the advent of hydrogen production and the creation of an offshore energy hub, the region is expected to see a significant increase in transmission capacity by 2050.

In 2050, projections indicate a notable increase in the total transmission capacity in the North Sea, largely driven by the integration of hydrogen production and the offshore energy hub. The increase in transmission capacity will far exceed today's grid, with

substantial growth coming from interconnections between wind farms and from wind farms to shore. The total transmission capacity will include connections between North Sea countries, such as from Germany to the Netherlands. Cable capacities to wind farms themselves will account for approximately 82 GW of the total capacity in the North Sea. Studies predict that the total transmission capacity could reach as much as 110 GW, reflecting the growing importance of the North Sea in Europe’s energy future.

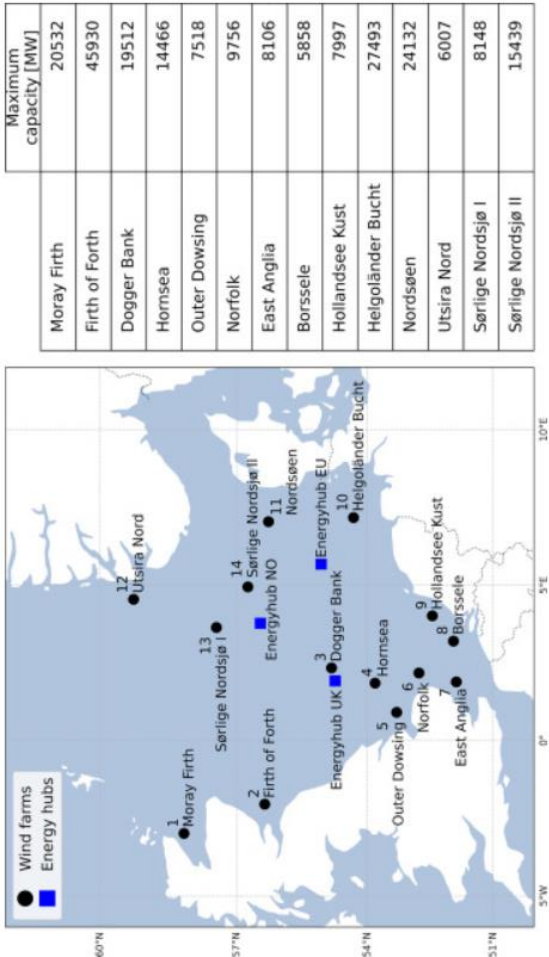
One important factor driving the expansion of transmission capacity is the introduction of the offshore energy hub. This hub allows for greater connectivity between offshore wind farms and countries along the European continent, enhancing the overall value of the North Sea’s wind resources. The offshore energy hub will increase the value of offshore wind power by facilitating

energy transmission between wind farms and surrounding countries, including Germany, the Netherlands, the UK, Norway, and Denmark. The flexibility provided by this energy hub makes it possible to efficiently transfer wind power and electricity, making the European power system more robust.

In addition to facilitating wind power transmission, the introduction of hydrogen production further enhances the energy grid. Hydrogen production helps to valorize excess wind power that would otherwise be curtailed, reducing curtailment from 24.9% to 17.1%. The combination of hydrogen production and the energy hub leads to a more significant reduction in curtailment, down to 9.6%. This highlights the potential of hydrogen to not only reduce waste but also play a crucial role in balancing energy demand and supply, ensuring that power generated from renewable sources is more efficiently utilized.

However, despite these improvements, the modelling suggests that there will still be some curtailment even when both hydrogen production and the energy hub are enabled. This indicates that while hydrogen production significantly reduces the amount of wasted wind power, there may still be limitations due to the incomplete modelling of renewable energy usage. Additionally, the production of green hydrogen will not primarily rely on curtailed renewable generation but will need to be supported by significant investments in renewable energy generation and transmission infrastructure.

Another important aspect of the North Sea’s energy future is its impact on power prices across Europe. As the demand for hydrogen increases, electrolyzers will consume large amounts of electricity from the power



grid, leading to changes in power prices. The electrolyzers will primarily consume cheap electricity during low-price seasons, such as summer, driving up prices. However, during peak demand periods in the winter, electrolyzers can help stabilize the power market by reducing prices through the increased use of the grid's generation and transmission capacity. This stabilizing effect is made possible by investments in grid infrastructure that support both hydrogen production and power transmission.

While the hydrogen economy has a stabilizing effect on power prices, it is not without challenges. In Norway, for example, the annual average expected power price increases significantly due to the high utilization of electrolyzer capacity, which drives up demand for electricity. In contrast, Germany and France see a slight reduction in their yearly average power prices, primarily due to the integration of Norwegian hydropower and offshore wind into the continental European grid. This highlights how the expansion of transmission capacity across borders plays a key role in balancing power supply and demand across regions.

Ultimately, the North Sea serves as a critical case study for the future of offshore hydrogen production. The integration of offshore wind farms, hydrogen production, and energy hubs will significantly impact Europe's power system, reducing curtailment, stabilizing power prices, and supporting the transition to a decarbonized energy grid. The large-scale deployment of hydrogen production will require substantial investments in renewable power generation and transmission infrastructure, but the benefits, including reduced curtailment and improved energy market stability, make it a crucial component of Europe's energy future. (Durakovic, et al., 2023)

China- Study Case Analyse

China's green hydrogen production has been rapidly expanding in recent years, with the Chinese government prioritizing the development of renewable energy sources and hydrogen technologies as part of its strategy to combat pollution and reduce carbon emissions. The country, which has already established itself as a global leader in the production of wind and solar energy, is moving towards expanding its production of Green Hydrogen with several pilot projects and facilities under development.

1. *China's largest offshore solar-hydrogen farm*

An important milestone is the Rudong integrated photovoltaic-hydrogen-storage project, which began operations in December 2024. Spanning 287 hectares and with a capacity of 400 MW, it combines solar energy, hydrogen production, and energy storage, generating 468 million kWh of electricity annually. It is anticipated to lower annual carbon dioxide emissions by approximately 309,400 tonnes, sulfur dioxide emissions by 562.6 tonnes, and nitrogen dioxide emissions by 1,125.3 tonnes, marking a major step in China's clean energy transition. (*China's largest offshore solar-hydrogen farm starts operation - Global Times*)



Based on these findings, China has the potential to supply large quantities of green hydrogen to Japan at competitive prices, contributing to Japan's energy transition and energy security. (Song et al., 2021)

Brazil- Study Case Analyse

Brazil's potential for offshore green hydrogen production is closely linked to its extensive coastline and abundant renewable energy resources. The country possesses an estimated technical capacity of approximately 700 GW for offshore wind energy in areas with depths of up to 50 meters, as identified by the Energy Research Company (EPE). The country's strategic position in the green hydrogen market is reinforced by its well-established renewable energy sector. With a large share of its electricity matrix already supplied by hydropower, Brazil has a favorable foundation for integrating offshore wind and solar power into hydrogen production. This potential aligns with the concept of "Powershoring," where countries leverage abundant local renewable energy to produce green hydrogen both for domestic use and export.

1. Economic Projections and Investment Opportunities

Economic projections highlight the scale of investment opportunities. The Brazilian Green Hydrogen Industry Association (ABIHV) estimates that the sector could attract up to \$7 trillion in investments by 2050, generating substantial tax revenues and employment. A study by McKinsey & Company suggests that Brazil could generate between \$15 and \$20 billion in hydrogen-related revenues by 2040, particularly in industries such as steel production, heavy transport, and petrochemicals. (Lara & Ritcher,2023)

2. Petrobras' Role in Green Hydrogen Development

Petrobras, a global leader in advanced technology for deep and ultra-deepwater

oil exploration and the leading energy company in Brazil, has been actively exploring pathways for green hydrogen development. The company has announced research initiatives and pilot projects focused on integrating renewable energy sources, such as offshore wind, into hydrogen production. Petrobras is also engaging in partnerships with international companies to advance technology and infrastructure for large-scale hydrogen production. These efforts are part of its broader strategy to transition towards a low-carbon energy portfolio while maintaining competitiveness in the global energy market.

Despite these advantages, Brazil still faces challenges in unlocking its full offshore hydrogen potential. Regulatory frameworks must be solidified to provide stability for investors, while infrastructure for large-scale hydrogen storage and transport needs to be developed. Addressing these hurdles will be crucial in positioning Brazil as a major global supplier of green hydrogen. As the demand for clean energy continues to grow worldwide, Brazil stands at the forefront of the transition, combining its renewable energy resources with new technologies to support the expansion of the hydrogen economy. (Lara & Ritcher,2023)

3. Environmental and Social Impacts in Ceará

In the specific case of Ceará, there are significant environmental concerns due to the installation of offshore wind farms for green hydrogen production. The region has ecologically sensitive areas, such as the Abrolhos Bank, where the installation of infrastructure could impact coral reefs and marine species breeding grounds, which are vital to local biodiversity. These impacts could be worsened by the construction and operation of wind farms, which could disturb the marine ecosystem and affect

local fishing, a traditional and important activity for coastal communities. Furthermore, fishing communities may be directly affected by the alteration of navigation routes and the presence of wind farm infrastructure, which could hinder access to their traditional working areas. There is also concern about the social impact of these changes on local activities, as many families depend on fishing for their livelihood. Evidence already exists of negative impacts from wind energy production in the region, such as disturbances in the marine ecosystem and local economies. This makes it essential to conduct in-depth environmental and social impact studies so that green hydrogen projects in Ceará can be implemented in a sustainable and balanced manner. (Lara & Ritcher,2023)

4. Ministry of Mines and Energy Study on the Potential of Green Hydrogen in Brazil

The Ministry of Mines and Energy, in partnership with the Brazil-Germany Cooperation for Sustainable Development (GIZ), conducted a study on the potential of green hydrogen in Brazil and the economic impacts of this transition. The study indicated that green hydrogen production in Brazil has the potential to generate a significant increase in the country's Gross Domestic Product (GDP), with an estimated growth of up to R\$ 61.5 billion by 2050. This GDP increase would be mainly driven by the integration of renewable energies, such as wind and solar, into hydrogen production, particularly in the Northeast regions, such as Ceará.

The research also highlights the positive impact that the transition to green hydrogen production could have on job creation, especially in Northeast regions, which have significant potential for utilizing wind and solar resources. These jobs could include both the construction

of renewable energy infrastructure, such as wind and solar farms, as well as the operation and maintenance of these facilities, and the formation of new jobs in industries associated with hydrogen production.

Furthermore, the study emphasizes the importance of effective public policies and investments in technology to ensure the success of Brazil's energy transition. The implementation of appropriate infrastructure for hydrogen storage and transportation will be crucial to enable the export of green hydrogen and to strengthen Brazil's position in the global hydrogen market, particularly in markets like Europe and Asia, which are heavily investing in the energy transition. (Lara & Ritcher,2023)

CONCLUSION

Offshore hydrogen production offers a transformative opportunity for the global energy transition, providing a scalable and sustainable solution to decarbonize sectors that are challenging to electrify. The integration of offshore renewable resources, such as wind and solar, with hydrogen production promises a reliable, clean energy source with diverse applications. By using excess renewable electricity, offshore wind energy can be converted into green hydrogen, stored and transported without relying on long-distance electricity transmission infrastructure, addressing issues related to grid congestion and energy storage. Offshore hydrogen is becoming more

economically viable as renewable technologies advance, and while high initial costs and infrastructure requirements pose challenges, these are increasingly manageable as the technology matures.

When comparing offshore production to onshore methods, offshore renewable energy systems offer significant advantages. Offshore systems can harvest energy from larger, more stable resources, and have access to a greater area for energy generation. This makes them less prone to the limitations faced by onshore systems, such as grid congestion. Combining offshore wind turbines with hydrogen production could solve some of the challenges of electricity storage and long-distance transmission, creating a robust solution for generating clean hydrogen. Moreover, emerging hybrid systems, like combining floating solar with hydrogen production, have the potential to increase energy yield and stability. There is also the possibility of repurposing decommissioned oil and gas platforms for green hydrogen production, improving energy security and economic feasibility. (Azad et al.,2023)

The completion of detailed scenarios for offshore hydrogen production has provided a clearer understanding of its economic and technical feasibility. Scenario analyses, such as those conducted in Portugal, highlight the importance of infrastructure, cost considerations, and government policies in determining the success of these projects. Large-scale offshore wind-to-hydrogen systems (Scenario 2) have demonstrated economic competitiveness and the potential for significant returns, while small-scale projects (Scenario 1) face challenges related to cost-effectiveness and subsidy dependency. These findings emphasize the critical

role that government incentives, such as subsidies and tax policies, play in making small-scale projects viable, while large-scale systems can operate profitably without government support.

The economic potential of offshore hydrogen is also enhanced by its ability to create new industries and jobs. By positioning countries with abundant renewable resources—like Portugal, Brazil, Norway, and China—as key players in the emerging global hydrogen market, offshore hydrogen production can contribute to economic growth, energy security, and the decarbonization of industries such as steel and transport. The growth of the green hydrogen market will also lead to the development of new revenue streams, such as oxygen sales, which can further enhance profitability.

However, while the economic benefits are substantial, the environmental and social impacts must not be overlooked. Areas like Ceará, Brazil, which feature ecologically sensitive regions, such as coral reefs and marine biodiversity hotspots, require careful planning to ensure that offshore hydrogen projects do not negatively impact local ecosystems or communities. As with any large-scale energy project, balancing the need for clean energy with the protection of the environment and local economies is crucial.

In conclusion, while offshore hydrogen production is still in its early stages, it holds enormous potential as a cornerstone of the global energy transition. With the right policies, strategic investments, and technological advancements, offshore hydrogen can become a pivotal element in reducing global carbon emissions and providing a sustainable energy future. As nations continue to explore this emerging sector, the integration of renewable energy

sources and hydrogen production, along with a focus on minimizing environmental and social impacts, will ensure that offshore hydrogen can meet the energy needs of tomorrow while fostering economic growth and sustainability.

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